

GrowBrain

User Manual



Cognitive Rehabilitation and Assistive
Android Game

This user manual provides guidance on the use of GrowBrain, an assistive Android game designed to support cognitive development in students with mild cognitive impairments.

Overview



GrowBrain
is an Android-based assistive
cognitive rehabilitation game designed for
students with Mild Cognitive Impairments (MCI).

It focuses on improving four core cognitive skills:
Memory, Attention, Logic, and Verbal Skills.
The system is intended to support SPED teachers
and school administrators while providing students
with an engaging, structured, and accessible
learning experience.

This manual explains exactly how each user role
uses the system, what each feature does, and how
tasks are performed step by step.

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INTRODUCTION

Overview

GrowBrain is an Android-based assistive cognitive rehabilitation game designed for students with Mild Cognitive Impairments (MCI). It focuses on improving four core cognitive skills: Memory, Attention, Logic, and Verbal Skills. The system is intended to support SPED teachers and school administrators while providing students with an engaging, structured, and accessible learning experience.

This manual explains exactly how each user role uses the system, what each feature does, and how tasks are performed step by step

SCOPE AND LIMITATIONS

Scope

GrowBrain is designed as an assistive cognitive rehabilitation tool for students with Mild Cognitive Impairments (MCI). The system focuses on improving four core cognitive domains: Memory, Attention, Logic, and Verbal Skills through game-based activities. It is intended to support SPED teachers and school administrators in monitoring student progress and adjusting learning activities.

Limitations

GrowBrain does not provide medical diagnosis, psychological assessment, or clinical treatment. The system does not cover physical, emotional, or behavioral therapy activities. GrowBrain is designed exclusively for Android devices and requires teacher supervision during student use.

System Requirements

2.1 Hardware Requirements

- Android smartphone or tablet
- Minimum of 2GB RAM
- Touchscreen interface

2.2 Software Requirements

- Android OS version 8.0 or higher
- Stable internet connection (for syncing data and reports)

2.3 User Accounts

- Administrator account
 - Teacher account
 - Student account (created by Admin or Teacher)
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User Roles and Access Levels

3.1 Administrator

The Administrator has full control over the system, including:

- Creating and managing teacher accounts
- Creating and managing student profiles
- Viewing overall system analytics
- Generating performance and usage reports

3.2 Teacher

Teachers are responsible for:

- Managing assigned student profiles
- Assigning cognitive games
- Adjusting game difficulty levels
- Monitoring student progress and performance

3.3 Student

Students can:

- Access assigned cognitive games
- Play games based on teacher-defined difficulty
- Receive real-time feedback during gameplay

Data Privacy and Ethical Use

GrowBrain handles student data strictly for educational and monitoring purposes. Access to student information is limited to authorized administrators and assigned teachers only. Student performance records are used to track cognitive progress and support instructional decisions. The system follows ethical guidelines to ensure confidentiality and responsible handling of data, especially since the users include minors and students with special educational needs.

GETTING STARTED

A. STUDENT USER GUIDELINES (TEACHER-GUIDED)

Important Note:

Students may not be able to read or understand written instructions. Teachers are required to provide verbal guidance and supervision when students use the GrowBrain application.

A.1 Preparing the Student for Use

- The teacher explains the purpose of the activity to the student in simple terms.
- The teacher demonstrates how to interact with the device (tapping, dragging, selecting).
- The teacher ensures the student is comfortable and attentive before starting.

A.2 Accessing the Game

- The teacher opens the GrowBrain application.
- The teacher selects the correct student profile.
- The teacher selects the assigned game and prepares the session.
- The device is handed to the student once the game is ready.

A.3 Playing the Game

- The student follows the instructions provided verbally by the teacher.
- The student interacts with the game elements as demonstrated.
- The application provides immediate visual and audio feedback.
- The student continues until the game session is completed or stopped by the teacher.

Student User

A. STUDENT USER GUIDELINES (TEACHER-GUIDED)

Important Note:

Students may not be able to read or understand written instructions. Teachers are required to provide verbal guidance and supervision when students use the GrowBrain application.

A.4 Teacher Supervision During Gameplay

- The teacher observes the student's interaction and response.
- Assistance is provided if the student shows difficulty or confusion.
- The teacher may pause or end the session when necessary.

A.5 Ending the Session

- The teacher ends the game session.
- The teacher reassures the student and provides positive feedback.
- The teacher returns the device to the system dashboard.

Teacher User

B. TEACHER USER GUIDELINES

B.1 Logging In

- Open the GrowBrain application.
- Enter the registered teacher username and password.
- Access the “For Teachers” to access the teacher dashboard.

B.2 Viewing Assigned Students

- From the dashboard, select My Students.
- View the list of assigned student profiles.
- Select a student to view personal records.

B.3 Assigning Cognitive Games

- Select a student profile.
- Click Assign Games.
- Choose one or more cognitive categories:
 - Memory
 - Attention
 - Logic
 - Verbal
- Select specific games under each category.
- Save the game assignment.

B.4 Setting Game Difficulty

- While assigning a game, select the difficulty level.
- Adjust difficulty based on the student’s ability and progress.
- Save the selected difficulty settings.

Teacher User

B. TEACHER USER GUIDELINES

B.5 Starting a Game Session

- Select the student profile.
- Click Start Session.
- Explain the game to the student verbally.
- Hand the device to the student and supervise gameplay.

AI Recommendations

The system provides intelligent recommendations:

- Click “Bulb icon” button
- The system analyzes the student’s last 50 game records
- Identifies weak areas (domains with <70% average accuracy)
- Recommends games to improve weak areas
- Suggests maintaining balance with strong areas

How AI Recommendations Work: The system computes the student’s average accuracy in each cognitive domain. It identifies which domains need improvement and recommends games that focus on those areas. At the same time, the system maintains activities in domains where the student performs.

B.6 Monitoring Student Performance

- After the session, open Performance Tab.
- Review accuracy, completion time, and attempt records.
- Use the data to adjust future game assignments.

B.7 Ending the Session

- Click End Session.
- Ensure performance data is saved.
- Log out of the system if finished.

Administrator

C. ADMINISTRATOR USER GUIDELINES

C.1 Logging In

- Open the GrowBrain's Admin Website.
<https://growbrainadmin.wasmer.app/>
- Enter administrator credentials.
 - Username: admin
 - Password: 123456
- Access the admin dashboard.

C.2 Managing Teacher Accounts

- Navigate to User Management.
- Select Add Teacher.
- Enter teacher details (name, username, credentials).
- Save the teacher account.

C.3 Managing Student Accounts

- Navigate to Student Management.
- Select Add Student.
- Enter student details:
 - Name
 - Age
 - Cognitive classification
 - Assigned teacher
- Save the student profile.

C.4 Editing or Deleting Accounts

- Select a teacher or student profile.
- Choose Edit or Delete.
- Confirm changes.

Administrator

C. ADMINISTRATOR USER GUIDELINES

C.5 Viewing System Analytics

- Navigate to Reports and Analytics.
- View system usage statistics.
- Monitor overall student performance trends.

C.6 Generating Reports

- Select Generate Report.
- Choose report type (student, teacher, system).
- Export or view the report.

C.7 System Maintenance

- Monitor user activity and system status.
- Coordinate updates and maintenance when required.
- Ensure data integrity and system availability.

Troubleshooting and Error Handling

Issue	Possible Cause	Solution
Unable to log in	Incorrect username or password	Verify login credentials
Game not loading	Internet connection issue	Check internet connection or restart the app
Progress not saved	Application used offline	Reconnect to the internet to sync data
Application not responding	System or memory issue	Restart the application or device



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